About SwimSignal

SwimSignal is a free five-day pollution risk forecast for river and lake swim spots in England. It covers 105 spots, 38 of them designated bathing waters. It uses the water companies' live storm-overflow data, rainfall forecasts and the river network. It is a forecast, not a water test.

Swimmers use it, and so do paddlers, ghyll scramblers, sailors and anyone else who gets in the water. The level is the same for everyone; how wet you get is what differs. Clubs, activity centres, Scouts and Guides leaders, schools and event organisers have their own page and tools.

How it works

For each spot, SwimSignal traces the river network upstream and finds every monitored storm overflow up to 60 km upstream along it. It takes what those overflows are doing now from the water companies' live feeds. It forecasts how likely each one is to spill on each of the five days, from the rain forecast and the overflow's own spill record. It then allows for how long sewage takes to travel downstream, how fast bacteria die off and how much the river dilutes it, so an overflow 2 km upstream counts for more than one 40 km up a tributary. Its sewage spill risk for each day is a 0–100 score for sewage from upstream reaching the spot. It is a score, not a probability. On rivers it also gives an E. coli estimate: the chance that a water sample would have more than 900 E. coli per 100 ml. 900 is the figure the Environment Agency uses to rate an inland bathing water "sufficient", over four seasons of samples, not a pass or fail test of one sample. At designated bathing waters it adds the Environment Agency's rating and its sampler's latest algae check. A day's level (low, moderate, high or very high) is the worst of these: the sewage spill risk, the E. coli estimate from May to September, a "poor" rating and an algae check from the last 14 days.

Where the trace finds no monitored overflow upstream, or at a lake that the river network links to no monitored overflow, such as a flooded quarry, the level is "No monitored overflow found upstream". It is not "no risk". Such a spot gets no sewage spill risk and no E. coli estimate. The page gives the rain in the last two days, and says that other risks apply: algae, wildlife, runoff and bathers. A rating of sufficient or poor, or algae at the last check, sets the level instead.

Each overflow is named first in plain words, such as "Addingham sewage works overflow", made from the name its water company gives it. The company's own name is shown beside it, smaller, because it is the name to look the overflow up by. Those names use abbreviations: WwTW and STW are a wastewater (sewage) treatment works, as is WRC, a water recycling centre; CSO is a

combined sewer overflow and SSO a storm sewage overflow; PS is a pumping station and SPS a sewage pumping station. A name with none of these is shown as the company gives it.

What it cannot see

- The sewage spill risk covers monitored storm overflows only. It knows nothing about a particular farm, road, boat, misconnected drain or incident. On rivers, the E. coli estimate allows for rain washing in pollution from other sources, but only on average.
- It does not count the treated sewage that sewage works let into rivers every day, only their storm overflows.
- It does not count wildlife or other bathers. On a lake it does not use the wind, which decides where river water goes. The method page says what you can do instead.
- Dŵr Cymru Welsh Water's overflows in Wales are not in SwimSignal's data: there is no licence to reuse them yet, and SwimSignal is waiting for Dŵr Cymru's terms. Where a spot's river runs from Wales, its page says how much of the river above it is in Wales. Dŵr Cymru's overflows in England are forecast from rain and their yearly spill record, with no "right now".
- Its level does not include the Environment Agency's short-notice advice against bathing after pollution or an algal bloom. On a bathing water's page, opening "Today's EA advice" loads the Agency's own panel, which has it.
- It says nothing about currents, cold water or other dangers.
- A "low" reading means nothing the site looks at flags the day. It does not mean the water is clean, or that it is safe to swim there.

How good it is

Forecasts are scored afterwards against what the overflows actually did and against the Environment Agency's water samples. The results are published on the accuracy page, including where the forecast was wrong. The E. coli estimate is tested on rivers in the May to September sampling season only. Please take any figures from the accuracy page: they change as new results come in.

Who runs it

SwimSignal is a free, non-commercial project run by one person, Ethan Buckley, not a company. It is not run by, connected with or endorsed by the Environment Agency, any water company, Water UK or the Met Office. The code is open source under the MIT licence. See the terms of use.

Help test it this winter

A small group of swimmers and river groups is testing SwimSignal from 19 October 2026 to 28 February 2027, and reporting what helps and what is wrong. The site stays open to everyone meanwhile. If you are one of the group, or would like to be, the testing page says what the site claims, what it does not, what to look for and how to tell us.

Contact

Email hello@swimsignal.co.uk. To report a problem or ask for a spot to be added, you can also use the feedback form or the spot request form. These prepare a private email for you to review and send; no GitHub account is needed.

Data

Storm overflow data from the water companies via the National Storm Overflow Hub, and data from the Environment Agency, Ordnance Survey, Open-Meteo, the Met Office and OpenStreetMap. Sources, credits and licences.

The files the site is built from are public too: every spot's forecast, the storm overflows and the accuracy scores. Data files lists each one, what it holds and its licence.

Show a spot's forecast on your site

A club, a council or anyone else with a non-commercial website can show one spot's forecast on their own web page: its headline, its five days, the caveat, when it was issued, a link to the full forecast and the data credits. It updates with each forecast. Paste this into the page, with the spot's id in place of `wharfe-burnsall`:

```
<iframe src="https://swimsignal.co.uk/embed.html?spot=wharfe-burnsall"
title="Pollution risk forecast from SwimSignal" width="100%" height="800"
style="border:0; max-width:480px" loading="lazy"></iframe>
```

A spot's id is the last part of its page's address: swimsignal.co.uk/spot/wharfe-burnsall/ is `wharfe-burnsall`. The card fits a column from 320 pixels wide. On 4 October 2026 the tallest card was 783 pixels high in a column 320 pixels wide, and 578 at 480 pixels; a card taller than its frame scrolls inside it. See the card for River Wharfe, Burnsall.

The terms of use allow this on a non-commercial website, as long as the card is shown whole: with its caveat, its credits and its link back to SwimSignal. The forecasts use rain from Open-Meteo's free service, which is for non-commercial use only. Ask first about any other use: email hello@swimsignal.co.uk.

For an event or a venue, the organisers' page gives the forecast for the day of the event and the storm overflows upstream of a spot, and links a sign to print with a QR code and a live sign for a

screen, which is this card, larger, updating itself.

Taking a group into the water? For clubs, centres and leaders says what the levels mean for swimming, paddling, sailing, rowing and ghyll scrambling, and which tools help.

SwimSignal is a free, non-commercial pollution risk forecast for river and lake swim spots in England, run by Ethan Buckley. A forecast, not a water test: check the signs at the water before you swim.

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